StudyBuddy
The Visual Teacher

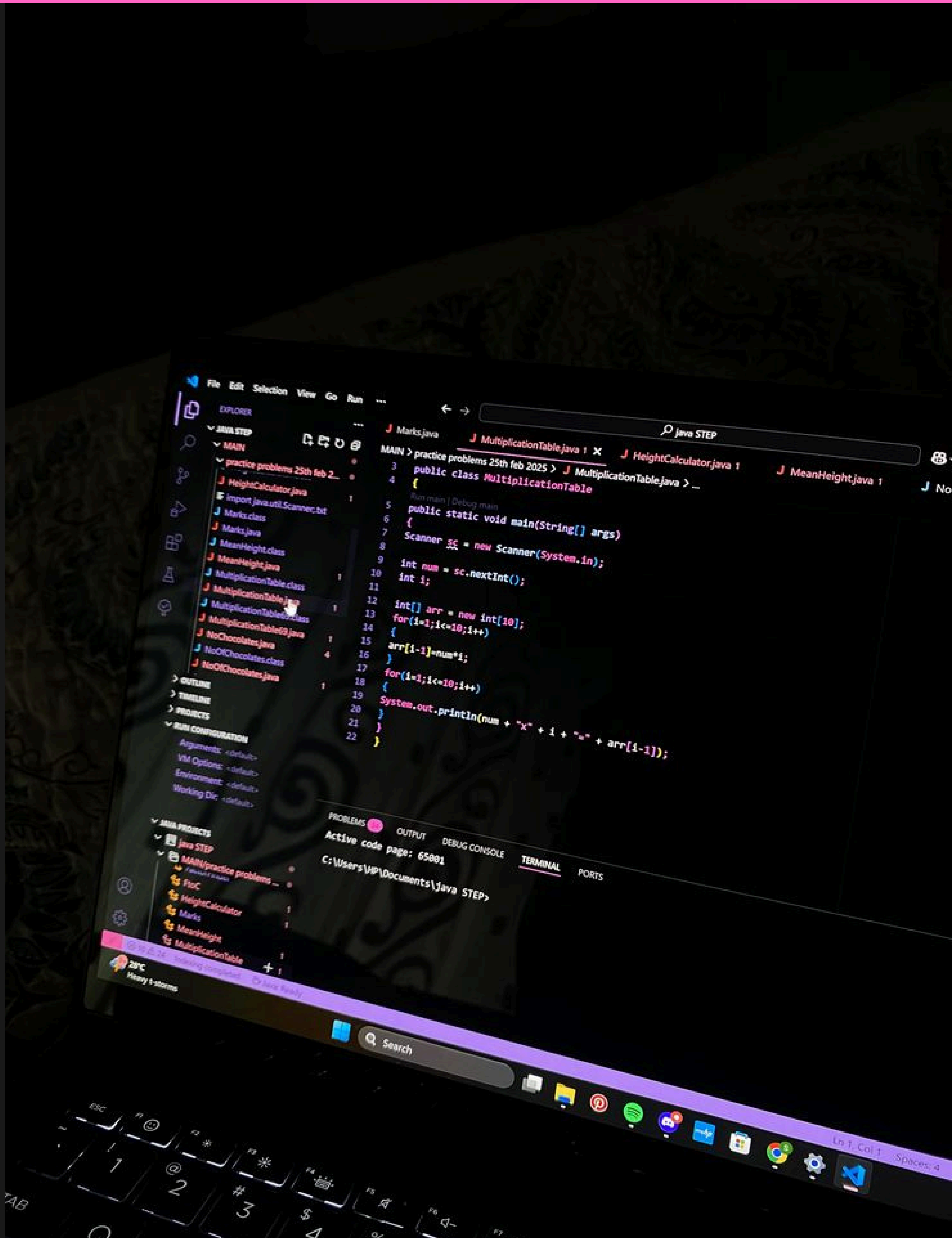
A simple and interactive virtual teacher designed to help beginners learn C++ concepts with ease.

Let's Begin

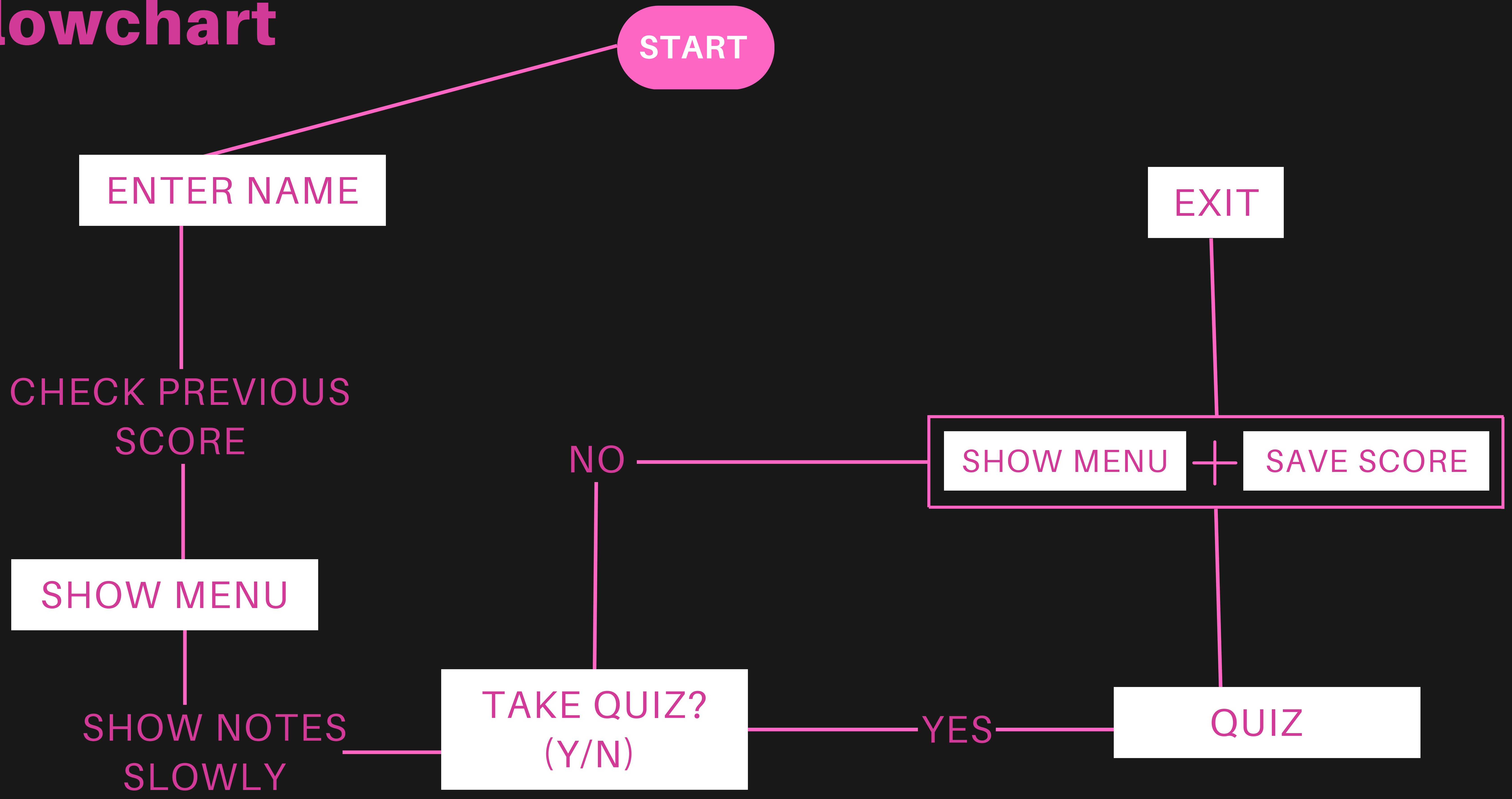
Presented by: **Anum Shahzad**

WHAT THIS PROGRAM DOES

- Displays topic notes from text files in an interactive, line-by-line style.
- Allows users to take a multiple-choice quiz for each topic.
- Loads quiz questions from topic-specific files automatically.
- Saves every quiz attempt to results.txt with name, topic, score, and date/time.
- Shows returning users their latest saved result when they enter their name again.



Flowchart


START
ENTER NAME
CHECK PREVIOUS SCORE
SHOW MENU
SHOW NOTES SLOWLY
TAKE QUIZ? (Y/N)
YES
QUIZ
SHOW MENU
SAVE SCORE
NO
EXIT

Notes System

We made a simple notes system. Each topic has a plain text file. We read it line-by-line using getline(), and then display each line slowly using our slowPrint() function. This makes the notes appear naturally instead of dumping all text fast.

- Reads from text file
- Shows line-by-line
- Uses slowPrint()
- Looks interactive

```
void showNotes(string topic) {

    string fileName;

    if (topic == "Variables") fileName = "variables.txt";
    else if (topic == "If Else Statements") fileName = "ifelse.txt";
    else if (topic == "Loops") fileName = "loops.txt";
    else if (topic == "Arrays") fileName = "arrays.txt";
    else if (topic == "Functions") fileName = "functions.txt";

    ifstream file(fileName);
    if (!file) {
        cout << "Error: Could not open " << fileName << endl;
        return;
    }

    cout << "\n------ Notes on " << topic << " ------\n";

    string line;
    while (getline(file, line)) {
        slowPrint(line);
    }

    file.close();
}
```

Name Recognition

When the user enters their name, we open the results file and search for any existing record that starts with their name. If we find one, we display their last quiz topic and score. This makes the program personalized.

- Enter name
- Program checks last record
- Greets user with old score

```
=============================================
         VIRTUAL TEACHER BOT
=============================================

Enter your name: anum

Welcome back anum!
Your previous quiz was on 'Functions' and you scored 2/5 on Fri Nov 28 23:26:24 2025

Hello anum! What would you like to learn today?
1. Variables
2. If Else Statements
3. Loops
4. Arrays
5. Functions
Enter your choice (1-5): _
```

File Handling

Files we use:

- Notes files — store topic explanations
- Quiz files — store the questions and answers
- Results file — stores the user's history

File handling is the backbone of this project.

We used:

- ifstream to read notes and quiz files
- ofstream to save scores
- ios::app append mode so new results get added without deleting old ones

Overview of All Functions

- waitDots()
- slowPrint()
- showNotes()
- takeQuiz()
- saveResult()
- getPreviousResult()
- showResult()
- main()

These are the eight functions that run the whole project.
Each one has a specific purpose, and together they build the complete StudyBuddy experience.

Function Explanations

- waitDots()
- slowPrint()
- showNotes()

waitDots()

Creates a loading animation using three dots. It gives the program a smooth, interactive feel.

slowPrint()

Prints each line with a short delay so the notes appear slowly, almost like reading from a real teacher.

showNotes()

Opens the correct text file for the selected topic, reads every line using getline(), and displays it using slowPrint().This keeps the notes clean and readable.

```cpp
// ------------------------------------------
// SIMPLE LOADING EFFECT
// ------------------------------------------
void waitDots() {
    for (int i = 0; i < 3; i++) {
        cout << ".";
        for (int j = 0; j < 100000000; j++);
    }
    cout << "\n";
}

// ------------------------------------------
// DELAY FOR NOTES
// ------------------------------------------
void slowPrint(string line) {
    cout << line << endl;

    // simple delay loop
    for (long i = 0; i < 200000000; i++);
}

// ------------------------------------------
// SHOW NOTES
// ------------------------------------------
void showNotes(string topic) {

    string fileName;

    if (topic == "Variables") fileName = "variables.txt";
    else if (topic == "If Else Statements") fileName = "ifelse.txt";
    else if (topic == "Loops") fileName = "loops.txt";
    else if (topic == "Arrays") fileName = "arrays.txt";
    else if (topic == "Functions") fileName = "functions.txt";

    ifstream file(fileName);
    if (!file) {
        cout << "Error: Could not open " << fileName << endl;
        return;
    }

    cout << "\n------ Notes on " << topic << " ------\n";

    string line;
    while (getline(file, line)) {
        slowPrint(line);
    }

    file.close();
}
```

TakeQuiz()

Loads quiz questions from the file, splits each line using a stringstream, stores them in a vector, shuffles them, and asks five random questions. At the end, it returns the final score.

```cpp
// ---------------------------------------------
int takeQuiz(string topic) {
    srand(time(0));
    vector<MCQ> questions;
    string fileName;

    if (topic == "Variables") fileName = "D:\\final project\\project_folder\\notes\\variablequiz.txt";
    else if (topic == "If Else Statements") fileName = "D:\\final project\\project_folder\\notes\\ifelsequiz.txt";
    else if (topic == "Loops") fileName = "D:\\final project\\project_folder\\notes\\loopsquiz.txt";
    else if (topic == "Arrays") fileName = "D:\\final project\\project_folder\\notes\\arraysquiz.txt";
    else if (topic == "Functions") fileName = "D:\\final project\\project_folder\\notes\\functionsquiz.txt";

    ifstream file(fileName);
    if (!file) {
        cout << "Error: Could not open " << fileName << endl;
        return 0;
    }

    cout << "\nStarting quiz on " << topic << " ";
    waitDots();

    string line;
    while (getline(file, line)) {
        stringstream ss(line);
        string q, a, b, c, correct;

        getline(ss, q, '|');
        getline(ss, a, '|');
        getline(ss, b, '|');
        getline(ss, c, '|');
        getline(ss, correct, '|');

        questions.push_back({q, a, b, c, correct});
    }

    file.close();
    random_shuffle(questions.begin(), questions.end());

    int score = 0;
    string ans;

    int total = min(5, (int)questions.size());

    for (int i = 0; i < total; i++) {
        cout << "\nQ" << i + 1 << ": " << questions[i].q << endl;
        cout << "A) " << questions[i].a << endl;
        cout << "B) " << questions[i].b << endl;
        cout << "C) " << questions[i].c << endl;

        cout << "> ";
        cin >> ans;

        if (ans == "a") ans = "A";
        if (ans == "b") ans = "B";
        if (ans == "c") ans = "C";

        if (ans == questions[i].correct)
            score++;
    }

    cout << "\n----------------------------------------\n";
    cout << "You scored " << score << " out of " << total << " in " << topic << "!\n";
    cout << "----------------------------------------\n";

    return score;
}
```

saveResult()

Writes the user’s name, topic, score, and date into results.txt using append mode. This makes the result permanent.

```cpp
// ------------------------------------------
// SAVE RESULT INTO A FILE
// ------------------------------------------
void saveResult(string name, string topic, int score) {
    ofstream fout("results.txt", ios::app);

    time_t now = time(0);
    string dt = ctime(&now);
    dt.pop_back();  // remove \n

    fout << name << "|" << topic << "|" << score << "|" << dt << "\n";
    fout.close();
}
```

getPreviousResult()

Scans the results file and finds the last saved record for that user. This allows us to greet returning users with their previous score.

```
// ---------------------------------------------
// READ PREVIOUS RESULT IF EXISTS
// ---------------------------------------------
string getPreviousResult(string name) {
    ifstream fin("results.txt");
    if (!fin) return "";

    string line;
    string last = "";

    while (getline(fin, line)) {
        if (line.find(name + "|") == 0) {
            last = line;
        }
    }

    fin.close();
    return last;
}
```

Function Explanations

- showResult()
- main()

showResult()

Displays a summary box that shows the user’s total score for the session.

main()

Controls the full program:

It welcomes the user, checks old scores, shows the topic menu, displays notes, runs quizzes, saves results, shows the sub-menu, and finally closes the program when the user decides to exit.

```cpp

// ------------------------------------------
// SHOW RESULT SUMMARY
// ------------------------------------------
void showResult(string name, int totalScore) {
    cout << "\n=========================================\n";
    cout << "                RESULT SUMMARY            \n";
    cout << "=========================================\n";
    cout << "Student Name : " << name << endl;
    cout << "Total Score  : " << totalScore << endl;
    cout << "=========================================\n";
}
```

```cpp
// ------------------------------------------
// MAIN PROGRAM
// ------------------------------------------
int main() {
    string name, topic, choice;
    int totalScore = 0;
    char takeQuizChoice;
    bool running = true;

    cout << "=========================================\n";
    cout << "         VIRTUAL TEACHER BOT             \n";
    cout << "=========================================\n\n";

    cout << "Enter your name: ";
    getline(cin, name);
    // CHECK PREVIOUS RECORD
    string last = getPreviousResult(name);
    if (last != "") {
        stringstream ss(last);
        string n, t, s, d;

        getline(ss, n, '|');
        getline(ss, t, '|');
        getline(ss, s, '|');
        getline(ss, d, '|');

        cout << "\nWelcome back " << name << "!\n";
        cout << "Your previous quiz was on '" << t << "' and you scored " << s << "/5 on " << d << "\n";
    }
    while (running) {
        cout << "\nHello " << name << "! What would you like to learn today?\n";
        cout << "1. Variables\n2. If Else Statements\n3. Loops\n4. Arrays\n5. Functions\n";
        cout << "Enter your choice (1-5): ";
        cin >> choice;

        if (choice == "1") topic = "Variables";
        else if (choice == "2") topic = "If Else Statements";
        else if (choice == "3") topic = "Loops";
        else if (choice == "4") topic = "Arrays";
        else if (choice == "5") topic = "Functions";
        else { cout << "Invalid choice!\n"; continue; }

        showNotes(topic);

```

Flow Chart

1
Ask for name and check previous record

2
Show Topic notes line by line

3
Take quiz

4
Show result

5
Exit

Thank You

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